What is a ConnectBox?



ConnectBox is a self-contained, self-powered, Wi-Fi-based portable media distribution system. It allows users to create a full featured ‘digital library’ from which media (text, video, audio and applications) can be shared on mobile phones, tablets, laptops, desktops in an off-the-grid environment without depending on access to the internet or electricity. Any device with WiFi capability can wirelessly connect to a ConnectBox and access/download content with a simple web browser. ConnectBox will support over 16+ simultaneous 360p video streams/download sessions from different devices. More devices can be connected if some devices are reading text or looking at media already downloaded. ConnectBox comes in several models – battery-powered, line-powered and with replaceable batteries – with US or EU chargers; see the ConnectBox Specification V10 for the details of each model.

Where would a ConnectBox be used?

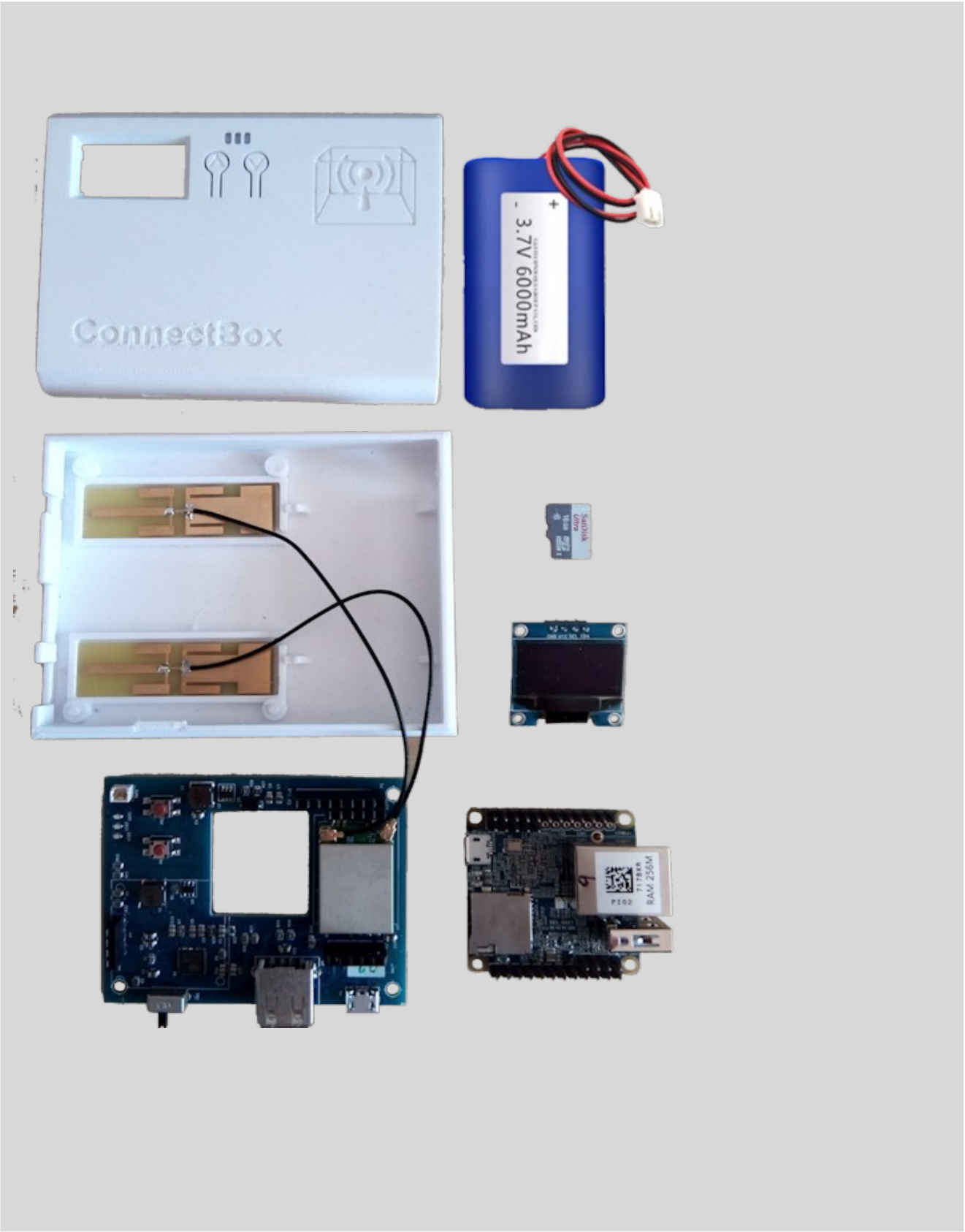
A ConnectBox acts as a local digital access point that isn’t capable of being restricted in traditional ways; allowing information to penetrate areas that would otherwise be unreachable. From educational, book study, language learning, audio programs, video content, text materials and applications can be delivered regardless of local internet connectivity or regional/national censorship. If you can provide power occasionally, a ConnectBox can provide the digital resources for learners. While internet access in many communities is unreliable or unaffordable, you can deliver important educational, healthcare and community resources without requiring users to

incur additional costs; they connect with the devices they already have. A ConnectBox can connect to an intermittent internet source either by phone WiFi hotspot or by bringing it into an area where WiFi internet is available. When the internet is available, the ConnectBox connects back to a central control center, which collects statistics on usage and media and offers new content packages that can be downloaded from the box's administration page. This control center can be deployed by organizations to manage all their ConnectBoxes in different areas. This provides a single point of view of all ConnectBox activity in the field. The control center software and its setup instructions are on GitHub (ConnectBox/connectbox-chathost).

The user interface is multi-lingual, provides thumbnail images of video and PDF content and allows custom images for any type of content. This also allows for groupings of media such as multiple audio files that belong to a single structure. This gives a single point of entry for a group of media that are numerically structured. To support multi-lingual capabilities at the root level of the storage media, folders are created with the ISO language codes. All media for that language is contained within that folder. A language selector at the top right corner of the screen allows you to change languages. The initial language is that of the user's phone if it exists on the device, otherwise the device's default language; a language the user picks with the selector is remembered and always comes first. Menus and interface text appear in the selected language, including right-to-left languages such as Arabic and Farsi. The menu wording for the 50 most spoken languages is built in, so it works without internet, and the Wi-Fi welcome page that opens when a phone joins is shown in the phone's language too.

What is it made of?

A ConnectBox is an open source project that uses off-the-shelf commercial hardware, a custom PCB, custom printed or molded case, a micro-SD flash drive, and software that when combined give you a small, low-powered webserver. The webserver acts as a captive portal and delivers files that are stored on the internal micro-SD flash drive or an external USB flash drive. It is based on freely available open source software that was developed by a number of different individuals and/or teams of people. It is available with build, connection, configuration and customization instructions for creating and customizing your very own ConnectBox. It runs a small Linux operating system on the hardware that can be tuned with added features and functions.


ConnectBox
3.7V 6000mAh
SanDisk
Ultra
RAM 256M

So how much does it cost?

The amazing thing about a ConnectBox is that it can be built for around $70 in hardware cost depending on the size of the storage space desired. Anywhere from 4GB to 512GB of storage space can be provided. With 64GB of storage space holding some 2,100 hours of audio or 270 hours of low-resolution MP4 video or 96 hours of standard MP4 video content.

So, what's so unique about a ConnectBox?

- Battery Powered – no need for AC mains, power adapters, external batteries. It's self-contained. Or non-battery line powered version is available.
- Its main menus are simple Icons which you can customize for your use. Additionally, more complex webpages can be developed and inserted as your requirements become more sophisticated.
- It has large storage; capacity limited only by the size of flash drive attached. It contains on board storage space as well as a USB port allowing additional flash drives to be attached.
- No internet service is required for it to be used. No wires – it is based on creating a WiFi hotspot. It fits into your shirt pocket.
- No special applications are required on the devices connecting to it. Any device with WiFi and a built-in web browser will work.
- The user interface dynamically scales to the screen size of the device attached.
- Built-in Administration interface allows for customization of the SSID, host name, WiFi channel and introductory text.
- When the ConnectBox itself has internet (by cable or Wi-Fi), the administrator can choose to share it with the people connected; it is off unless turned on.
- The Administration interface asks for its password every time and logs out as soon as you leave it, so no phone stays logged in as administrator.
- You can share any type of files easily.
- Streams/downloads movies, music, audio files, photos, text files and device applications.
- Built in statistics for users who have accessed the device and files downloaded/viewed.
- Can be recharged by a simple 5V USB charger (USB-C or micro-USB, depending on the model) or added external solar panels.
- It can be branded to your own organization's liking.
- Offline websites (ZIM files: Wikipedia, Stack Exchange, PhET, TED and more) open in a built-in reader with search.
- Every web page and offline website has a home button back to the ConnectBox menu.
- Word (.docx and older .doc), Excel (.xlsx, .xls) and PowerPoint (.pptx and older .ppt) files open as easy-to-read pages on any phone (presentations with their speaker notes), with the original file available to download.
- Download buttons show the size of each file before it is downloaded.
- While new content is being loaded, visitors see a progress page in their own language instead of an empty menu.

ConnectBox Specifications:

Storage: Internal micro-flash memory – ext4 format (from 4GB to 256GB capacity)

External USB flash memory – FAT32, exFAT, NTFS or ext4, up to 512GB

LED’s: Green-Power on, Red-Battery charging, Orange – WiFi Operational
Switches: Power On/Off, Forward & Backward push buttons

Power, battery life, charging time and dimensions depend on the model (P, R, S or J): see the ConnectBox Specification V10.

Wireless Standards: IEEE 802.11n (b/g/n)

Frequency: 2.4-2.4835 GHz, Access Point Mode, operating channels 1 - 11, -28dBm at source

Support 64/128 bit WPA-PSK/WPA2-PSK

Range: Typically 50m (160’) line-of-sight

Signal Rate: Up to 300 Mbps

Operating:

Temp 15C – 40C, Humidity 0% ~75% non-condensing

Storage:

Temp -20C – 50C, Humidity 0%~75% non-condensing

Typical Storage Media Usage (depending on encoding & resolution):

Low-resolution MP4 videos (176×144 pixel) ~500 kb/s
MP4 format videos (adaptable) ~1.4 Mb/s
MP3 Audio format ~64 Kb/s
1-5 page optimized PDF ~200 KB / PDF E book ~1 – 5 MB / book
Photo ~3.2 MB/ photo
MP3 Song ~3.5 MB / song
5-min Web video ~20 – 30 MB / video 5-min
HD video ~300 MB / video